Problem 1. Express each of these system specifications using predicates, quantifiers and logical connectives. Define any propositional functions that you use, and choose predicates that allow as many concepts as possible to be represented in the logic.

a) The e-mail address of every user can be retrieved whenever the archive contains at least one message.
b) At least one console must be accessible during every type of fault condition.
c) There are at least two paths connecting every two distinct endpoints on the network.

Answer.

The domains are unrestricted for all these answers.

a) Predicates:
   - $U(u) \equiv$ "u is a user"
   - $R(u) \equiv$ "the e-mail address of user u can be retrieved"
   - $A(a) \equiv$ "a is an archive"
   - $M(m, a) \equiv$ "m is a message from archive a"

   Proposition: $\forall x \forall y \left((U(x) \land A(y)) \to \exists z \left(M(z, y) \to R(x) \right) \right)$
b) Predicates:
   - $C(c) \equiv$ "c is a console"
   - $F(f) \equiv$ "f is a fault condition"
   - $A(f, c) \equiv$ "console c is accessible during fault f"

   Proposition: $\forall x \left(F(x) \to \exists y \left(C(y) \land A(x, y) \right) \right)$
c) Predicates:
   - $P(p) \equiv$ "p is an endpoint"
   - $R(r) \equiv$ "r is a path"
   - $C(r, a, b) \equiv$ "path r connects point a and point b"

   Proposition: $\forall p_1 \forall p_2 \left((P(p_1) \land P(p_2) \land \neg(p_1 \equiv p_2)) \to \exists r_1 \exists r_2 \left(R(r_1) \land R(r_2) \land \neg(r_1 \equiv r_2) \land C(r_1, p_1, p_2) \land C(r_2, p_1, p_2) \right) \right)$

Problem 2. Assume the domain consists of all CS 441 students. Translate these statements using quantifiers.

a) Lina has best performance in the class.
   Let $B(x, y)$ represent the statement "x performs in class better than y."
b) Only one student was nominated by the class to solve a recitation problem on the board.
   Let $N(x, y)$ represent the statement, "x nominated y to solve a recitation problem on the board".

Answer.

a) $\forall x\,(\neg(x \equiv \text{Lina}) \to B(\text{Lina}, x))$
b) Predicate: $P(x) \equiv$ "x is presenting on the board."
   Propopoposition: $\forall x\,(\neg(x \equiv P(x)) \to N(x, P(x)))$

Problem 3. Let $S(x) \equiv$ "x is a CS 441 student", $T(x) \equiv$ "x is a CS 441 TA", $E(x,y) \equiv$ "x enjoys CS 441 more than y", and $A(x,y) \equiv$ "x attends recitation more than y". Assume the domain for each of these propositional functions consists of all the students at Pitt. Translate the logical expressions into *natural* English sentences.

a) $\exists x\,(S(x) \land \forall y\,(T(y) \to A(x,y)))$
b) $\exists x\,(T(x) \land \forall y\,((T(y) \land x \neq y) \to E(x,y)))$

Answer.

a) There exists a CS441 student that attends recitation more than all the CS441 TAs.
b) There exists a CS441 TA that enjoys CS441 more than all the other TAs.

Problem 4. Use rules of inference to show that the hypotheses imply the conclusion. Consider the domains and propositions given.

a) Hypotheses: "It rained today"; "If it rained today, the festival will not occur"; and "If the festival does not occur, I am sad". Conclusion: "I am sad".
Let $r \equiv$ "It rained today", $f \equiv$ "The festival occurs", $s \equiv$ "I am sad".

b) Hypotheses: "The farm will not make profit"; "If the chickens lay eggs and the price of car insurance doesn't increase, then the eggs are sold and the farm makes profit"; "Either the chickens lay eggs or a new car will be bought". Conclusion: "The price of car insurance will increase or a new car will be bought".
Let $c \equiv$ "The chickens lay eggs", $e \equiv$ "Eggs will be sold", $p \equiv$ "The farm will make profit", $n \equiv$ "A new car will be bought", $t \equiv$ "The price of car insurance will increase".

c) Hypotheses: "Somebody in CS 441 likes grilled cheese sandwiches," "All people are either lactose intolerant or likes ice cream," and "Every person who is lactose intolerant does not like grilled cheese sandwiches." Conclusion: "There is a person in CS 441 who likes ice cream."
Let $C(x) \equiv$ "x is a student in CS 441", $G(x) \equiv$ "x likes grilled cheese sandwiches", $L(x) \equiv$ "x is lactose intolerant", $R(x) \equiv$ "x likes ice cream". Assume all domains are "all people".

Answer.

a) Given: r, $r \to \neg f$, and $\neg f \to s$, with the conclusion s. Using hypothetical syllogism, we can get that $r \to s$. So, by modus ponens, if it rained today, then I am sad.

b) Given: $\neg p$, $(c \land \neg t) \to (e \land p)$, and $c \lor n$, with the conclusion $t \lor n$. First use contraposition to get that $(\neg e \lor \neg p) \to (\neg c \lor t)$. Given $\neg p$, we can get $(\neg c \lor t)$. Given that $c \lor n$, we can get $t \lor n$ using resolution.

c) Given: $\exists x\,(C(x) \land G(x))$, $\forall x\,(L(x) \lor R(x))$, $\forall x\,(L(x) \to \neg G(x)))$, with the conslusion $\exists x\,(C(x) \land R(x))$. We first use existential instantiation to conclude that $C(a) \land G(a)$ for an element a, and then since $G(a)$ is true, we can conclude that $\neg L(a)$ is true by contraposition. Finally, since $\neg L(a)$ is true, the second statement is also true since is an disjunction, which means that they also like ice cream.

Problem 5. Determine whether each of these arguments is valid. If an argument is correct, what rule of inference is being used?

a) If n is a real number such that $n > 1$, then $n^2 > 1$. Suppose that $n^2 > 1$, then $n > 1$.
b) If n is a real number with $n > 3$, then $n^2 > 9$. Suppose that $n^2 \leq 9$, then $n \leq 3$.
c) If n is a real number with $n > 2$, then $n^2 > 4$. Suppose that $n \leq 2$, then $n^2 \leq 4$.

Answer.

a) Hypothesis: $\forall x \in \mathbb{R}\left((n > 1) \to (n^2 > 1)\right)$
Conclusion: $\forall x \in \mathbb{R}\left((n^2 > 1) \to (n > 1)\right)$
Incorrect, $(p \to q) \nleftrightarrow (q \to p)$ (the converse is not logically equivalent).
b) Hypothesis: $\forall n \in \mathbb{R}\left((n > 3) \to (n^2 > 9)\right)$
Conclusion: $\forall n \in \mathbb{R}\left((n^2 \leq 9) \to (n \leq 3)\right)$
Correct, modus tollens. The conclusion is also just a contrapositive of the original statement.
c) Hypothesis: $\forall n \in \mathbb{R}\big((n > 2) \to (n^2 > 4)\big)$
Conclusion: $\forall n \in \mathbb{R}\big((n \leq 2) \to (n^2 \leq 4)\big)$
Incorrect, the inverse of the proposition is not logically equivalent to the original statement.